ZHAW · VISUALISATION & DATA SCIENCE STORYTELLING II · FS 2025/26

Optimising Product Placement Using Grocery Purchase Data

Turning 3.2 million Instacart orders into assortment, cross-sell and retention decisions for grocery delivery.

Lara Pestrin · Jonathan Kull · Veprim Mehmeti

THE CHALLENGE

Grocery delivery lives or dies by its assortment

Thousands of products.
Weekly stocking decisions.
Today — mostly intuition.

THE QUESTIONS WE SET OUT TO ANSWER

1. Which products are purchased most frequently?
2. Which products are commonly bought together?
3. Which products earn the highest customer loyalty?
4. When — by day and hour — are orders placed?
5. What does this mean for assortment and retention?

THE DATA

Built on 3.2 million real grocery orders

3.2M
Orders analysed

206K
Unique customers

49.7K
Unique products

21
Departments, 134 aisles

The public Instacart 2017 dataset — six linked tables, fully anonymised.

WHO IT'S FOR

Two people make these calls every week

Sandra

ASSORTMENT PLANNER

34 · Business Administration · 5 years in grocery retail · no coding

NEEDS

A fast visual overview of what sells — and what sells together.

PAIN

Slow, manual reports; co-purchase patterns are invisible.

Marco

BUYER

28 · Retail Management · 2 years in e-commerce grocery · no coding

NEEDS

Evidence for listing, delisting and supplier negotiations.

PAIN

Decisions lean on gut feel; loyal niche products stay hidden.

THE PRODUCT

An interactive Streamlit dashboard

Built for exploration, not static reporting.

01
KPI Overview

- Orders, customers, products
- Reorder rate & peak hour
- Top department

02
Product Development

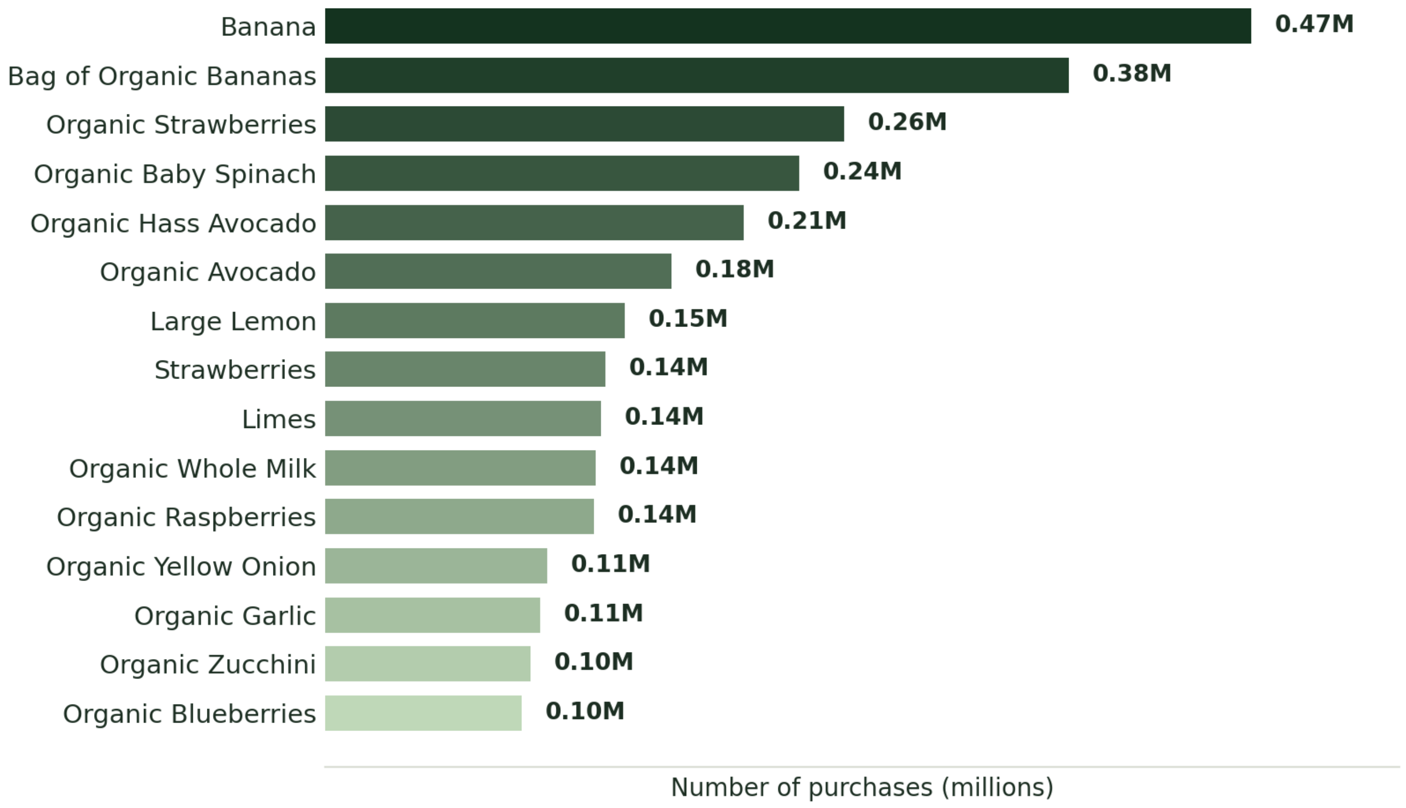
- Top sellers
- Department sales
- Products bought together

03
Customer Retention

- Reorder rate
- Hidden gems
- Shopping-time heatmap

Run it locally: streamlit run deployment/app.py

Fresh produce owns the top of the basket

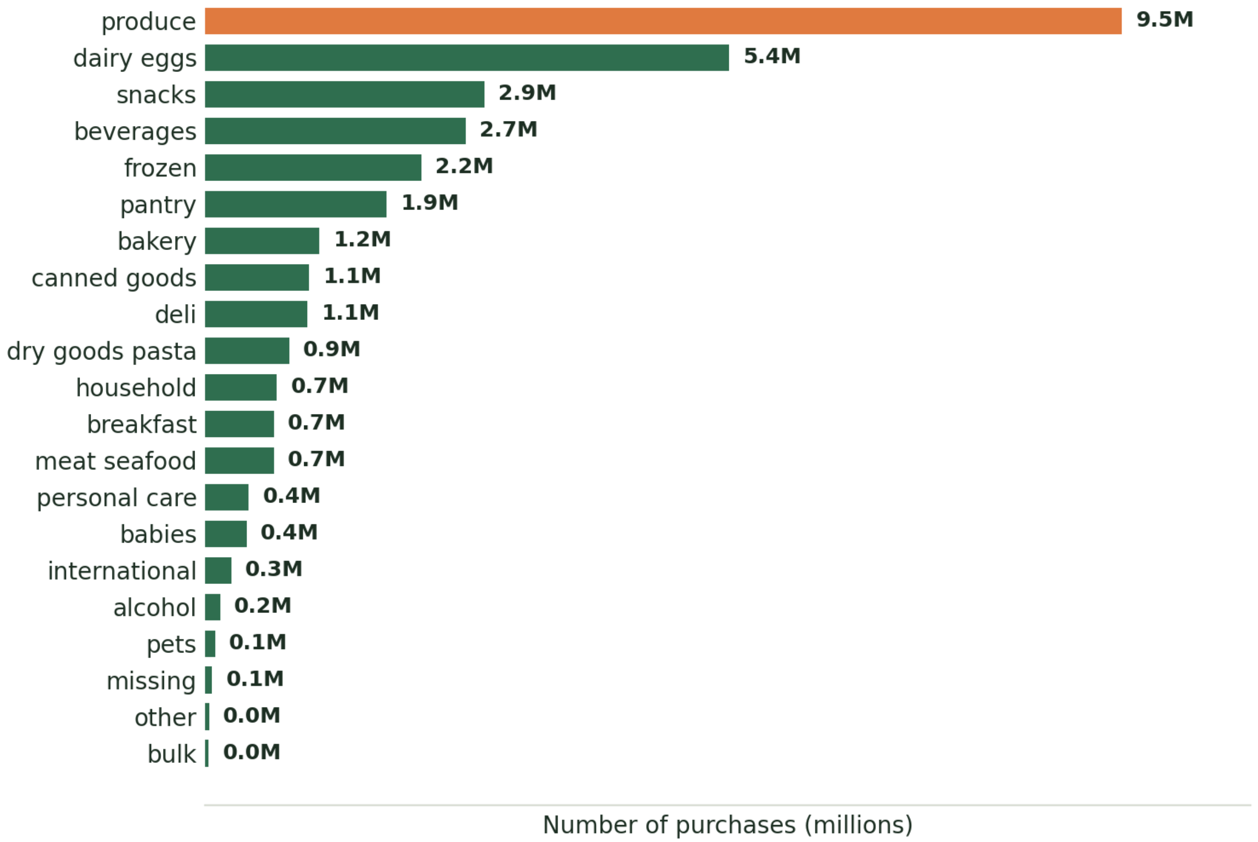


WHAT WE FOUND

Every one of the top 15 products is fresh fruit or vegetables.

RESULT 02 · PRODUCT DEVELOPMENT

Produce drives almost a third of all volume

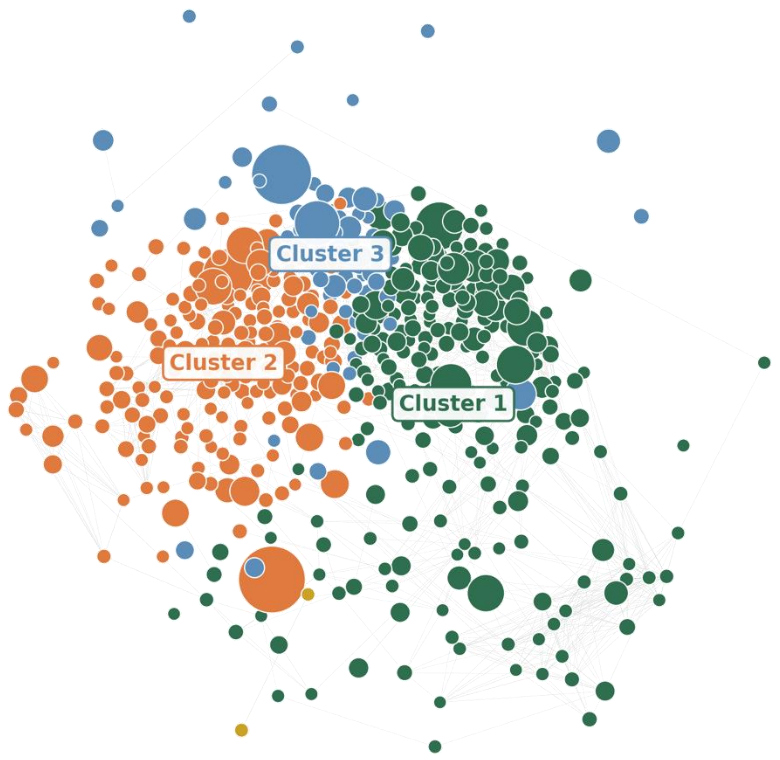


WHAT WE FOUND

Produce and Dairy & Eggs dwarf every other department.

RESULT 03 · PRODUCT DEVELOPMENT

Co-purchase analysis reveals natural baskets

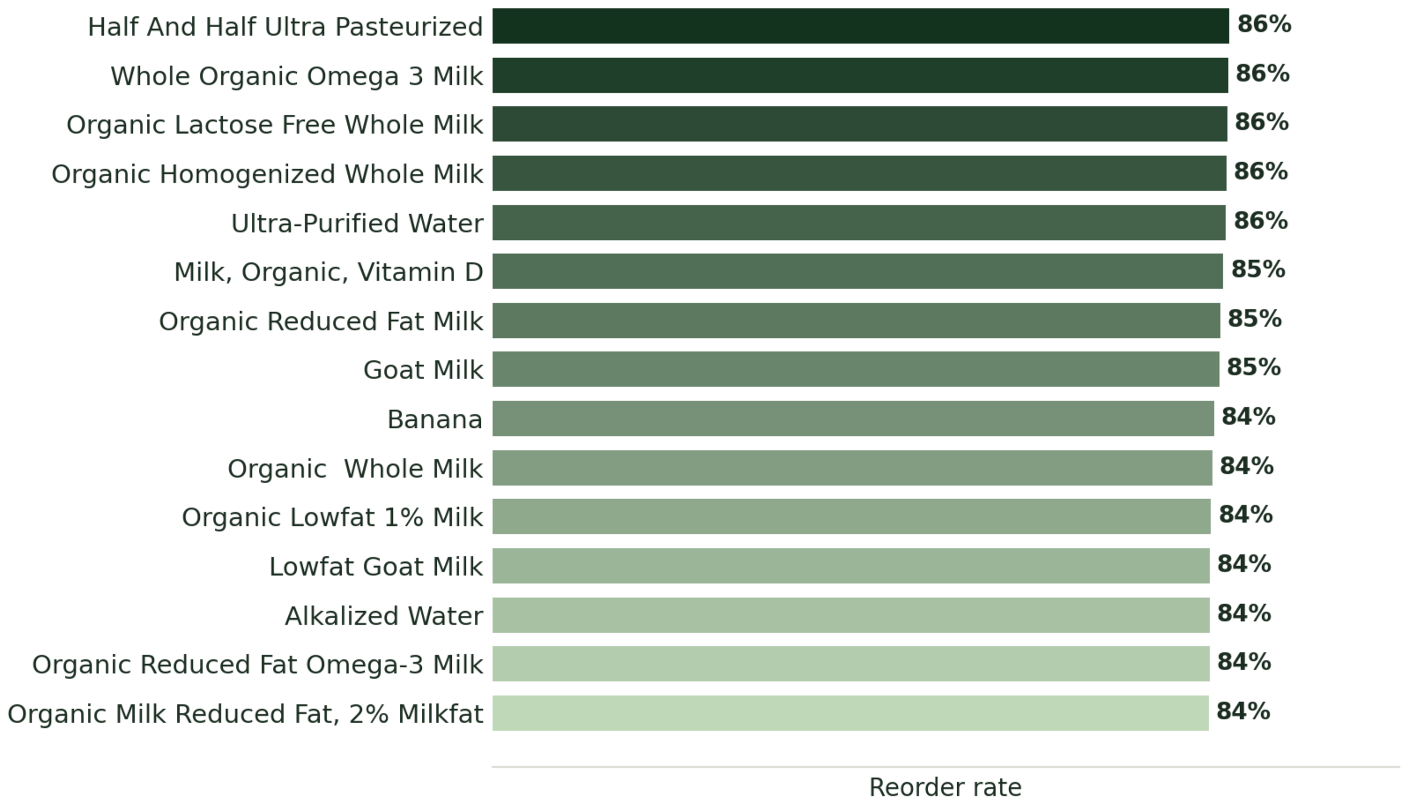


WHAT WE FOUND

Co-purchase patterns fall into three natural baskets.

- **Fresh vegetables & aromatics**
  spinach, avocado, onion, garlic, lemon
- **Everyday fruit & milk**
  banana, berries, apples, whole milk
- **The organic basket**
  organic bananas, berries, almond milk

Customer loyalty is built on milk

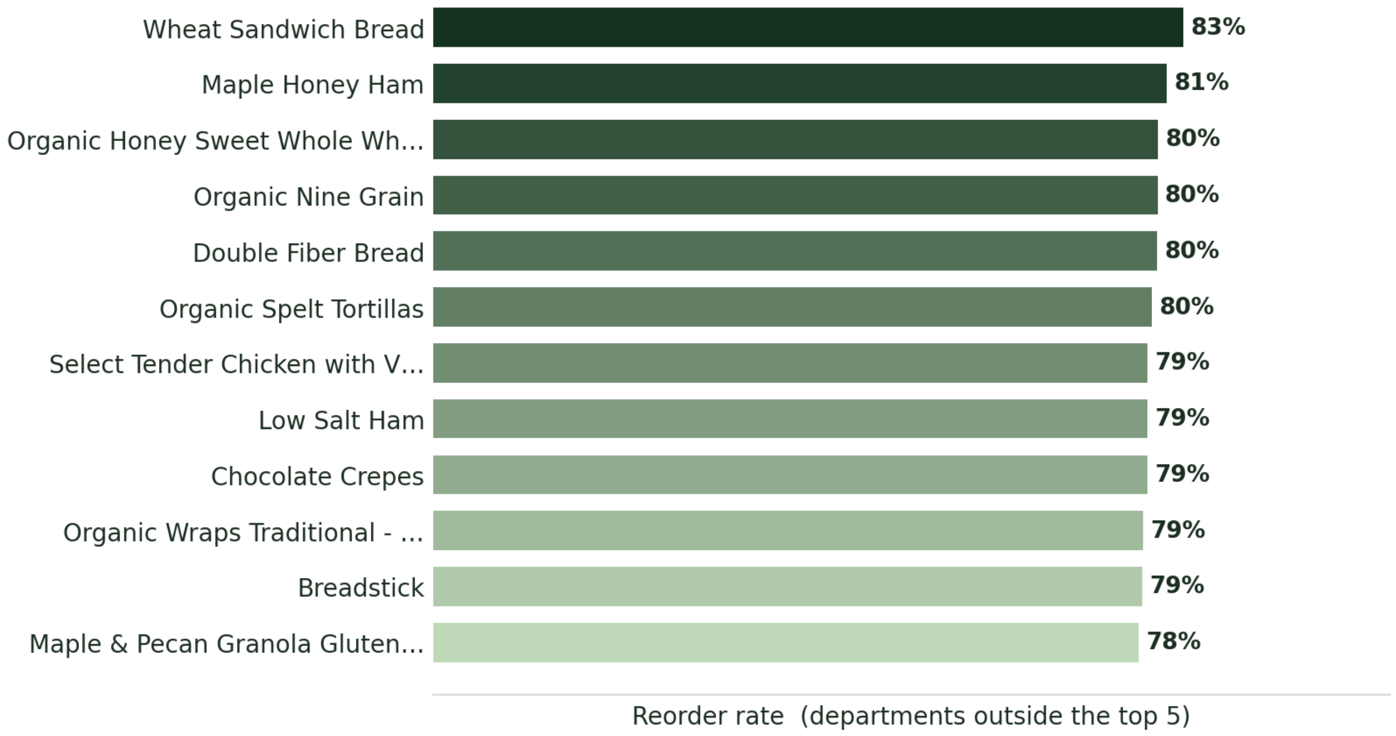


WHAT WE FOUND

Loyalty runs on milk and water — 84 to 86% reorder.

RESULT 05 · CUSTOMER RETENTION

Hidden gems: small categories, loyal fans

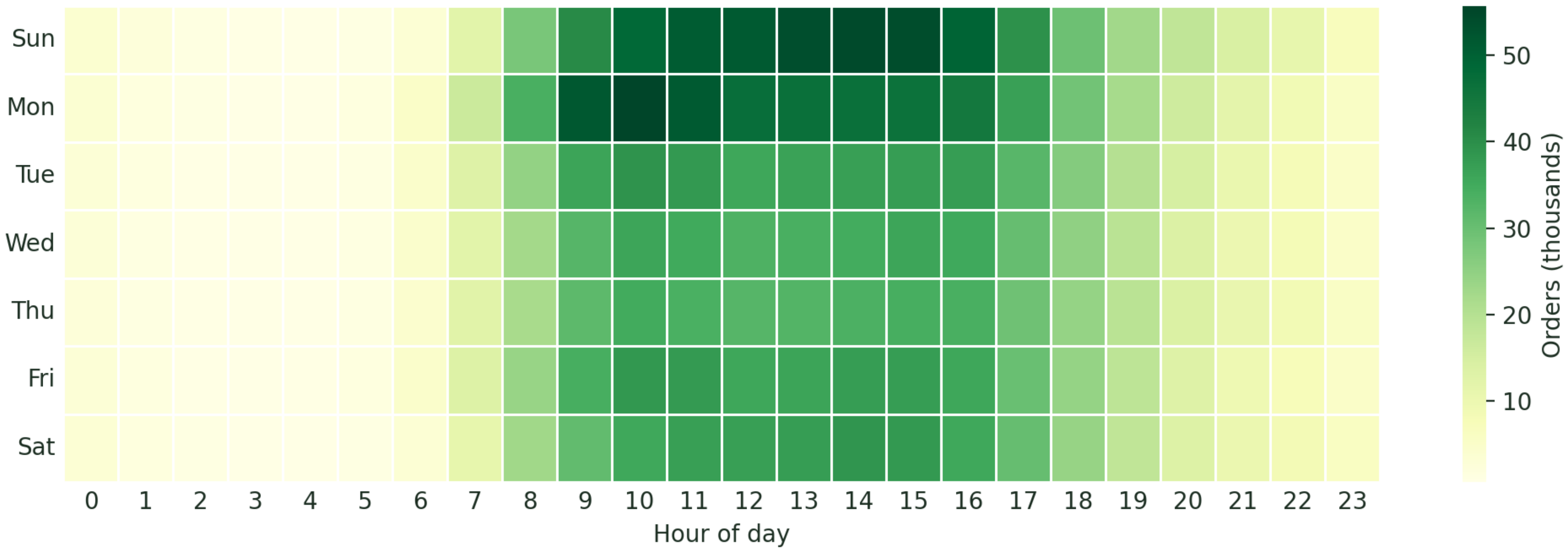


WHAT WE FOUND

Niche breads and deli items keep 78 to 83% of their buyers.

RESULT 06 · CUSTOMER RETENTION

Sunday and Monday mornings carry the week



WHAT WE FOUND

Orders cluster on Sunday and Monday, roughly 9:00 to 16:00 — and fall away sharply outside daytime hours.

WHAT IT MEANS

From data to assortment decisions

01

Build on fresh

Produce and organic are the core.

FOR SANDRA

02

Bundle the clusters

Group and cross-sell what sells together.

FOR MARCO

03

Subscribe the staples

Milk and water — auto-reorder offers.

FOR SANDRA

04

Protect the gems

Keep loyal niche products on the shelf.

FOR MARCO

THANK YOU

Three million orders, one clear shelf strategy.

Anonymised purchase data, turned into decisions Sandra and Marco can act on — without writing a line of code.

Repository github.com/peslar01/ad24-11-grocery-product-placement-analysis

Run it streamlit run deployment/app.py

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